

**“VISVESVARAYA TECHNOLOGICAL UNIVERSITY,
BELAGAVI”**



**B.L.D.E.A's V.P. DR. P.G. HALAKATTI COLLEGE OF
ENGINEERING AND TECHNOLOGY, VIJAYAPUR --
586103**

**DEPARTMENT OF COMPUTER SCIENCE
& ENGINEERING**

Artificial Intelligence Assignment Project

Course Code : BCS515B

Name : Pavan Sabani

USN : 2BL22CS149

Name : Sanjana Baragi

USN : 2BL22CS158

UNDER THE GUIDANCE

Mrs.D.M.Ijeri

HOD

Dr . LEENA RAGHA

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INDEX

Sl.no	Title	Page No
1	Introduction	04
2	Dataset	05
3	Objectives	06
4	Logistic Regression	06
5	AI Model Selection	07
6	Standard Libraries	07
7	Architecture of Model	08
8	Statistical Overview	09
9	Implementation & Accuracy	10
10	Output of the Model	11
11	Conclusion	12

INTRODUCTION

Diabetes is a chronic metabolic disorder characterized by elevated blood sugar levels due to insufficient insulin production, improper insulin utilization, or both. According to the World Health Organization (WHO), diabetes affects millions globally, posing a significant public health challenge. If left undiagnosed or poorly managed, it can lead to severe complications, including cardiovascular diseases, kidney failure, nerve damage, and vision loss.

Current situation of diabetes detection :

- The current state of diabetes detection has significantly advanced, yet challenges remain.
- Traditional methods of diabetes diagnosis include measuring fasting blood glucose levels, oral glucose tolerance tests (OGTT), and HbA1c tests, which provide reliable metrics for assessing blood sugar control over time.
- While these methods are accurate, they often require clinical visits, are invasive, and may not be easily accessible in low-resource settings.
- As a result, many cases, especially in developing regions, remain undiagnosed until complications arise.
- Despite these advancements, global disparities in healthcare access and affordability persist.
- Rural and underserved populations face barriers to regular screening and care.
- Efforts are underway to bridge these gaps through mobile health applications, telemedicine, and point-of-care diagnostics, making diabetes detection more accessible.
- However, raising awareness about diabetes and promoting routine screenings remain critical to reducing the global burden of undiagnosed cases

DATASET : PIMA INDIA

This dataset is originally from the National Institute of Diabetes and Digestive and Kidney Diseases. The objective of the dataset is to diagnostically predict whether or not a patient has diabetes, based on certain diagnostic measurements included in the dataset. Several constraints were placed on the selection of these instances from a larger database. In particular, all patients here are females at least 21 years old of Pima Indian heritage. The size of the dataset is relatively small by modern standards, which could lead to overfitting in complex models.

Features :

- Pregnancies: Number of pregnancies.
- Glucose: Plasma glucose concentration over 2 hours.
- BloodPressure: Diastolic blood pressure (mm Hg).
- SkinThickness: Triceps skinfold thickness (mm).
- Insulin: 2-hour serum insulin (μ U/ml).
- BMI: Body Mass Index ($\text{weight in kg}/(\text{height in m})^2$).
- DiabetesPedigreeFunction: A function that scores the likelihood of diabetes based on family history.
- Age: Age in years.
- Outcome: Target variable (1 if the patient has diabetes, 0 otherwise).
- The target variable is binary:
 - 1: Indicates the presence of diabetes.
 - 0: Indicates the absence of diabetes.

Characteristics:

- Contains **768** entries.
- Some features have missing values represented as zeroes, which are handled by replacing them with the column mean in this project.

Applications :

The Pima Indian Diabetes Dataset serves as a benchmark in:

- **Machine Learning:** Used to test classification algorithms such as logistic regression, decision trees, support vector machines, and neural networks.
- **Feature Engineering:** Provides a platform to explore data preprocessing, handling missing values, and feature scaling.
- **AI in Healthcare:** Helps researchers develop predictive models for diabetes risk assessment.

OBJECTIVES

Statement :

The primary aim of this project is to predict the likelihood of diabetes in individuals by analyzing various health metrics and characteristics. These metrics include glucose levels, blood pressure, body mass index (BMI), insulin levels, and, for females, the number of pregnancies. By developing a predictive machine learning model, the project seeks to facilitate the early detection of diabetes risk, empowering individuals and healthcare providers to take timely preventive or management measures.

Clarity of Problem Statement

- The project defines a clear goal: to predict diabetes risk using a machine learning model.
- The focus is on health data-driven predictions, where input features like glucose levels, BMI, and blood pressure are used to evaluate the likelihood of diabetes.

ALGORITHM : Logistic Regression

Logistic regression is a widely used supervised learning algorithm in machine learning, especially for binary classification problems like diabetes detection. It predicts the probability of an outcome (e.g., presence or absence of diabetes) based on input features and is highly interpretable, making it a popular choice in healthcare applications.

Steps in Using Logistic Regression for Diabetes Detection:

1. Data Preprocessing:

- Handle missing or anomalous values in the dataset.
- Normalize features to ensure faster convergence.

2. Model Training:

- Fit the logistic regression model to the training data.

3. Evaluation:

- Use metrics like accuracy, precision, recall, F1-score, and AUC-ROC to evaluate the model.

4. Prediction:

- Input unseen data into the trained model to predict diabetes risk.

AI MODEL SELECTION

1. Appropriate Choice of Model :

The project employs Logistic Regression, a suitable and well-understood method for binary classification tasks like diabetes prediction. Logistic regression is a popular model choice because of its interpretability and efficiency, especially with well-preprocessed datasets.

2. Justification of Model Selection :

Logistic Regression is appropriate for this problem due to the binary nature of the outcome (diabetes: yes/no) and the relatively straightforward relationship between the predictors and the target variable. It provides a baseline model, making it easier to interpret and analyze before moving to more complex models if needed.

3. Comparison with Alternatives :

While logistic regression is chosen for its simplicity, alternative models like Decision Trees, Random Forest, or Support Vector Machines could have been considered for potentially better performance. However, these models might require more computational resources and fine-tuning. The project could expand on these comparisons for a comprehensive analysis.

STANDARD LIBRARIES

1. NumPy (import numpy as np):

- Used for numerical computations and handling arrays, including replacing zeros with NaN values.

2. Pandas (import pandas as pd):

- Used for data manipulation and analysis, particularly for loading, summarizing, and cleaning the dataset.

3. Scikit-learn:

- `train_test_split`: Splits the dataset into training and testing subsets.
- `StandardScaler`: Standardizes features by removing the mean and scaling to unit variance.
- `LogisticRegression`: Implements logistic regression for classification.
- `accuracy_score`: Computes the classification accuracy.
- `confusion_matrix`: Evaluates the performance using a confusion matrix.
- `classification_report`: Provides a detailed classification report with precision, recall, and F1 score.

ARCHITECTURE OF MODEL

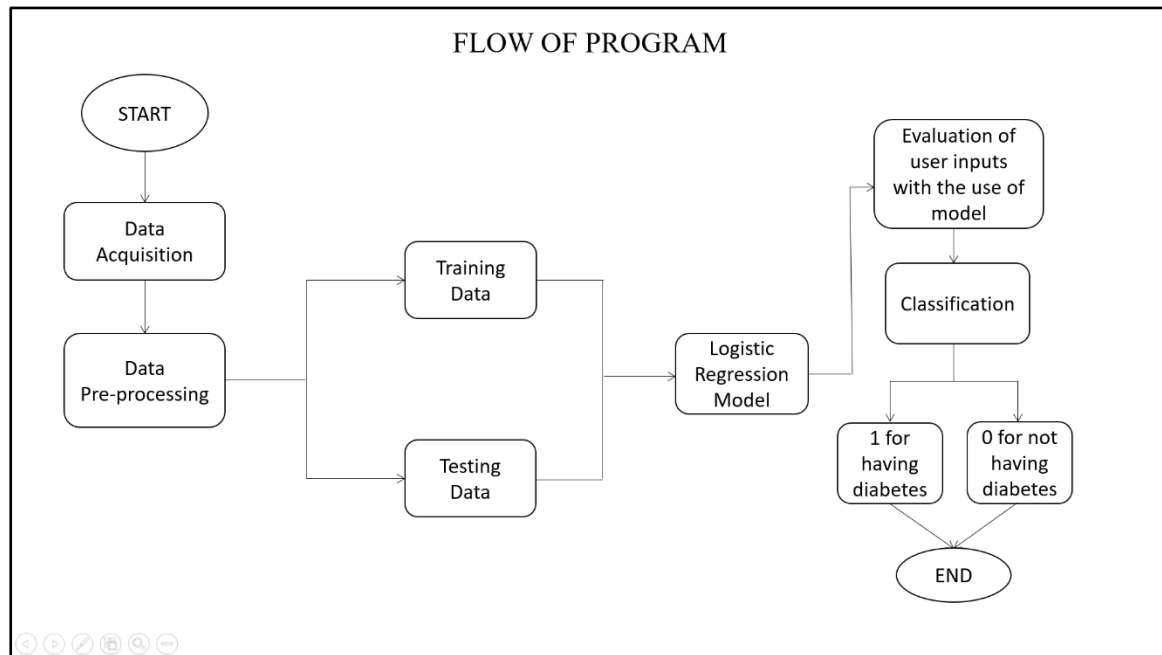


Fig 7.1 Flow chart of the Model

The provided figure 7.1 flowchart outlines the process of a machine learning program designed for diabetes prediction. It begins with the **Start** node, leading to the **Data Acquisition** step, where the dataset is collected. This is followed by **Data Pre-processing**, where the dataset undergoes cleaning and preparation, such as handling missing values or scaling the features. The data is then split into two subsets: **Training Data** and **Testing Data**. The **Training Data** is used to train the **Logistic Regression Model**, which forms the foundation of the prediction process. After training, the model is evaluated using **Testing Data** to ensure accuracy and reliability.

The model is then used to process **User Inputs**, which are evaluated using the trained logistic regression model. This produces an **Output** that predicts whether the user has a risk of diabetes. The result is binary: **1 for having diabetes** or **0 for not having diabetes**, ultimately concluding with the **End** node. This flowchart clearly visualizes the logical steps of data handling, model training, and prediction output in the machine learning pipeline.

STATISTICAL OVERVIEW

When the program is executed , it provides the following information on the model.

```
PS C:\Users\pavan> & C:/Users/pavan/AppData/Local/Programs/Python/Python312/python.exe "d:/skill_dev/simple projects/diabeatase detection/db_detection.py"
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

Fig 8.1 : Health data

Fig 8.1 showcases the **initial rows of the dataset** used for training and evaluating a diabetes prediction model. Each row represents a data instance containing multiple features that are critical for diagnosing diabetes. The last column, Outcome, represents the target variable, where 1 indicates diabetes presence and 0 indicates its absence.

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479	31.992578	0.471876	33.240885	0.348958
std	3.369578	31.972618	19.355807	15.952218	115.244002	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.078000	21.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000	27.300000	0.243750	24.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000	32.000000	0.372500	29.000000	0.000000
75%	6.000000	140.250000	80.000000	32.000000	127.250000	36.600000	0.626250	41.000000	1.000000
max	17.000000	199.000000	122.000000	99.000000	846.000000	67.100000	2.420000	81.000000	1.000000

Fig 8.2 : Statistical Summary of the Diabetes Dataset

Fig 8.2 presents the **statistical summary** of the dataset used for training and evaluating a diabetes prediction model. Each column represents a feature, and the rows show various statistical measures for these features. The total number of records are **768**. The average(mean) value for each feature of Glucose, BMI etc. The range of values for each feature i.e minimum and maximum values. Quartiles (25%, 50%, 75%) these indicate the data distribution(for example : 25% of values fall below the first quartile (e.g., Glucose = 99)).

Confusion Matrix:

[[82 17]
[21 34]]

Fig 8.3 : Model Evaluation

Fig 8.3 displays the **performance evaluation**. The confusion matrix provides the counts of correctly and incorrectly classified instances for the two classes. 82: True Negatives, 17: False Positives, 21: False Negatives, 34: True Positives

IMPLEMENTATION & ACCURACY

Implementation:

1. Data Preparation:

- Handling Missing Values: Replaced zeros in certain columns (Glucose, BloodPressure, SkinThickness, Insulin, and BMI) with NaN and then filled these missing values with the column mean using `df.fillna(df.mean(), inplace=True)`.
- Feature-Target Split: Features (X) were separated from the target variable (Outcome).
- Train-Test Split: The dataset was divided into training (80%) and testing (20%) subsets using `train_test_split`.

2. Feature Standardization:

- Used `StandardScaler` to standardize the features, ensuring that each feature had a mean of 0 and a standard deviation of 1. This step is crucial for models like Logistic Regression, which perform better with standardized data.

3. Model Training:

- Implemented a Logistic Regression model from `sklearn.linear_model`.
- The model was trained on the standardized training dataset using the `fit` method.

4. Model Prediction:

- Predictions were made on the testing dataset using the `predict` method.
- Additionally, user input data was scaled and predicted using the same model.

5. Performance Metrics:

- Confusion Matrix: Used to analyze the number of true positives, true negatives, false positives, and false negatives.

Accuracy:

- Score : **75.32%**
- Computed using `accuracy_score`, which calculates the percentage of correctly classified instances.

OUTPUTS OF THE MODEL

```
Enter your gender (male/female): female
Enter the number of pregnancies: 2
Enter glucose level: 184
Enter blood pressure level: 66
Enter skin thickness: 29
Enter insulin level: 0
Enter BMI: 24.3
Enter diabetes pedigree function: 0.627
Enter age: 25
The model predicts: You have a risk of diabetes.
```

Fig 11.1: Sample output of female patient

Fig 11.1 shows the output for a diabetes detection of female patient. It takes some health-data as input required to test diabetes of patient and based on the provided input values, the model predicts that the patient **do have risk of diabetes.**

Fig 11.2 shows the output of male patient. Based on the input values, he is not at immediate risk of diabetes. This likely means the glucose level, insulin level, and other indicators are within a range that **does not strongly suggest the presence of diabetes.**

```
Enter your gender (male/female): male
Enter glucose level: 85
Enter blood pressure level: 66
Enter skin thickness: 24
Enter insulin level: 0
Enter BMI: 28.3
Enter diabetes pedigree function: 0.332
Enter age: 29
The model predicts: You are not at risk of diabetes.
```

Fig 11.2: Sample output of male patient

```
Enter your gender (male/female): female
Enter the number of pregnancies: 2
Enter glucose level: 98
Enter blood pressure level: 82
Enter skin thickness: 19
Enter insulin level: 85
Enter BMI: 27.4
Enter diabetes pedigree function: 0.42
Enter age: 34
The model predicts: You are not at risk of diabetes.
```

Fig 11.3: Sample output of female patient

Fig 11.3 shows the output of female patient. This figure demonstrates that model evaluates multiple health metrics to assess the risk of diabetes and predicts that she is **not having the risk of diabetes.**

CONCLUSION

This project serves as a meaningful step toward integrating AI into healthcare diagnostics. It demonstrates the potential of machine learning to empower both patients and healthcare providers with tools for early disease detection, ultimately contributing to better health outcomes and reduced healthcare costs.

Moreover, this project emphasizes the critical role AI can play in transforming traditional healthcare practices. By providing data-driven insights, machine learning models like the one implemented here bridge the gap between raw clinical data and actionable predictions. This empowers healthcare providers to make informed decisions with greater accuracy and efficiency.

For patients, such tools mean quicker access to assessments and potentially earlier interventions. In regions with limited healthcare resources, AI-driven diagnostics can help democratize access to quality healthcare, addressing disparities and improving equity. Automated systems reduce the dependency on specialized personnel, making diagnostic services more scalable and cost-effective.